Bubble collapse

Is a sonoluminescing bubble a large-scale picture of the knot's implosion, and does its light carry the knot's topology? — CLAUDE.md §3h, register entry H30

kronspice simulation report · 2026-11-07 · argon R0 = 4.5 μm, 1.3 atm at 25.02 kHz; recorded run: run_output.txt

© 2026 fdamador@comcast.net

Summary

A single sonoluminescing argon bubble is put on the matched water flask of *acoustic_drive* as an equivalent circuit (the bubble returns to scope only as a cited equivalent-circuit model, CLAUDE.md §1 non-goals), and compared with the implosion of knotted light in *knotted_light* (H29, H30). The user's question (2026-11-05): is the energy at the centre of a collapsing bubble a large-scale picture of the knot's implosion, and does its light carry the knot's topology? The run follows every joule of one acoustic cycle, loads the flask's drive chain with the bubble, tests reversibility, estimates the flash and its spectrum against measurement, and measures the flash's circular polarization in the field solver.

- **One cycle, ledger closed.** 14.2923 nJ in from the sound field = 1.04039 nJ viscous heat + 13.2519 nJ radiated sound (Δ stored -4.99×10^{-18} J); residual 6.2×10^{-10} of the input. 91 % of the radiated sound leaves within ±10 ns of the minimum.
- **Circuit and Keller–Miksis agree on the slow motion** (R_{max} 32.60 against 32.73 μm, minimum at phase 0.5170 against 0.5181) and differ at the collapse (wall Mach 0.781 against 0.516).
- **The bubble is a small load on the flask:** 357.66 μW against 6.95727 mW into the flask's mode; the generator EMF for 1.3 atm rises from 1.8981 V to 1.9240 V.
- **Reversible only without losses:** the lossless bubble returns to $|\Delta R|/R = 2.5\times10^{-12}$; with sound radiation it misses by 0.90.
- **The flash estimate:** 12.7746 fJ, about 6.2×10^{4} photons, FWHM 180 ps, against measured 10^{5}–10^{7} photons and 60–250 ps; the width matches, the colour and intensity do not.
- **No topology in the flash:** the hopfion gives $P_c = +0.9996$ (exact +0.9999); the thermal core gives mean +0.0058, rms 0.0184 over six seeds, consistent with zero.

What is being tested

The idea (H30 context, user question). The knot's implosion and explosion (H30, *knotted_light*) might have a large-scale counterpart in the collapse of a sonoluminescing bubble, and the bubble's light might carry the knot's topology. **Standard prediction.** The collapse is a nonlinear, irreversible, spherically symmetric flow of matter stopped by gas pressure; its light is thermal (Hilgenfeldt, Grossmann & Lohse, *Nature* 398, 402, 1999; Brenner, Hilgenfeldt & Lohse, *Rev. Mod. Phys.* 74, 425, 2002), with degree of circular polarization $P_c = 0$ within counting noise. A knot of field energy is one helicity and gives P_c of order 1.

The model

The bubble as a circuit

In the acoustic domain the across variable is pressure and the through variable is volume velocity U = dV/dt. From the far liquid to the gas:

```
p∞(t) ── [ M(V) || R_rad(V) ] ── R_visc(V) ── wall ── gas p_g(V) − 2σ/R
```

Element	Value	Source
liquid inertance	M = ρ/(4πR), so ½MU² = 2πρR³Ṙ²	Rayleigh 1917
radiation resistance	$R_{rad} = \rho c/(4\pi R^2)$	pulsating sphere: M || $R_{rad} = (\rho c/4\pi R^2)\cdot jkR/(1+jkR)$, Kinsler et al. §7.2

© 2026 fdamador@comcast.net

Element	Value	Source
wall viscosity	R_{visc} = μ/(πR³) (= 4μṘ/R per U)	Newtonian liquid
gas	argon, adiabatic γ = 5/3, hard core R0/h = 8.86	Brenner, Hilgenfeldt & Lohse 2002
surface	2σ/R	IAPWS R1-76(2014) σ = 71.69 mN/m at 300 K

M depends on the volume, the circuit's own "charge", so the inertance is a Kron inductance on a coordinate. With momentum Π = M·U_M, dΠ/dt = Δp + ½(dM/dV)·U·U_M, where the ½M′ term is the Christoffel (speed-voltage) term of Kron 1951 (Epilogue eqs. 3–4). Without the two losses this is the Rayleigh–Plesset equation exactly. R_{rad} adds the liquid's compressibility to first order in Ṙ/c, the job Keller–Miksis does; Keller–Miksis runs beside it as the twin. Integration is Dormand–Prince 5(4) at tolerance 10^{-10}.

Parameters: argon bubble R0 = 4.5 μm in water at 300 K, drive 1.30 atm at 25.02 kHz (the matched flask's peak); ρ = 996.556 kg/m³, c = 1501.52 m/s (IAPWS-95), μ = 853.8 μPa·s (IAPWS R12-08), σ = 71.69 mN/m.

The ledger

It closes by construction (d(½Π²/M)/dt = U_M·Δp, and U_M·Δp = U·Δp − Δp²/R_{rad}):

```
∫ −p_ac·U dt = Δ[½MU_M² + p_g(V−Vh)/(γ−1) + 4πσR² + P0·V] + ∫R_visc·U² dt + ∫Δp²/R_rad dt
in from the sound field = Δ stored + viscous heat + sound radiated
```

Results

Part 1: one acoustic cycle (the sixth from rest)

	Circuit	Keller–Miksis
R_{max}	32.60 μm	32.73 μm
R_{min}	0.730 μm	0.769 μm
minimum at phase [cycles]	0.5170	0.5181
max wall Mach |Ṙ|/c	0.781	0.516

- Peak gas: 2.35×10^{9} Pa and 14985 K (adiabatic); R_{max}/R_{min} = 44.7.
- Ledger over the cycle: in from the sound field 14.2923 nJ = Δ stored -4.99483×10^{-18} J + viscous 1.04039 nJ + radiated sound 13.2519 nJ; residual 6.2×10^{-10} of the input. The collapse energy P0·V_{max} is 14.7069 nJ.
- 91 % of the radiated sound leaves within ±10 ns of the minimum: the collapse turns a 25 kHz input into a nanosecond outgoing pulse (peak 2.7 kW for a few nanoseconds, per the interactive report).
- Energy density goes from 3.86 J/m³ in the sound field to 9.03×10^{9} J/m³ in V_{min}, a concentration of 2.3×10^{9}.
- The two models agree on the slow quantities (R_{max} to 0.4 %, the minimum's phase to 0.1 %) and differ at the collapse, where Ṙ/c is no longer small: the circuit is first order in Ṙ/c, while Keller–Miksis keeps (1 − Ṙ/c) factors that slow the wall more. Neither is exact there (Brenner et al. discuss Gilmore and full gas dynamics).

Part 2: the bubble as a load on the drive chain

The bubble's fundamental admittance $Y_1 = -U_1/P_1 = 4.123\times10^{-14} + 2.825\times10^{-14}$j m³/(Pa·s) is put at node p of the matched water chain as a resistor and a capacitor. This harmonic balance is exact for power, because only the fundamental of U does work against the sinusoidal centre pressure: ½Re(Y_1)P_a² = 357.66 μW equals the cycle work × f = 357.66 μW.

	No bubble	With bubble
peak	25.025 kHz	25.024 kHz
pressure per volt	69.40 kPa/V	68.46 kPa/V
generator EMF for 1.3 atm	1.8981 V	1.9240 V
source power	26.1816 mW	26.5416 mW
into the liquid (mode)	6.95727 mW	6.95727 mW
into the bubble	—	357.66 μW
kronspice ledger residual	0.0	0.0

© 2026 fdamador@comcast.net

The bubble takes about 5 % of what the flask's own losses take. export/bubble_on_flask.cir is the chain with the bubble element (RBUBBLE, CBUBBLE), with its KiCad sheet and the MNA against torn results.

Part 3: reversibility

The knot's implosion is its explosion run backwards (Maxwell's equations are time-symmetric). Here the bubble runs from R_{max} through the collapse and back out (2 × the fall time); then its momentum is flipped and the drive run backwards.

Viscosity	Radiation	|ΔR|/R on return	$\|\Delta\Pi\|/\Pi_{max}$
off	off	2.5×10^{-12}	4.6×10^{-13}
on	off	6.5×10^{-2}	1.3×10^{-2}
off	on	9.0×10^{-1}	1.8×10^{-2}
on	on	8.0×10^{-1}	8.1×10^{-3}

Without losses the bubble is a conservative (Hamiltonian) system and retraces its collapse, like the knot. The radiated sound makes it irreversible, because the outgoing pulse does not come back; running a bubble backwards would need that pulse to converge on it again, the job H30's Huygens surface did for the knot.

Part 4: the light and its spectrum

Free–free (bremsstrahlung) emission from argon ionized per the Saha equation, assuming a uniform, optically thin core (Rybicki & Lightman eqs. 5.14b, 5.15b; Gaunt factor 1.2; NIST argon levels):

- energy per flash 12.7746 fJ (8.7×10^{-7} of the collapse energy), peak 63.831 μW, ionization 5.8×10^{-3} at the peak;
- flash FWHM 180 ps, about 6.2×10^{4} photons (energy / kT at the peak);
- measured: 10^{5}–10^{7} photons per flash (about 10^{-12} J), widths 60–250 ps (Gompf et al., *PRL* 79, 1405, 1997). The estimate is low by one to two orders in photons and right in width.

Quantity	Model	Measured	Agreement
Width, UV band (250–300 nm) / red band (600–700 nm)	164 / 176 ps	equal (Gompf et al. 1997)	yes
Spectral shape	peaks near 475 nm, falls into the UV	rises into the UV to water's edge near 200 nm	no
Planck-fit temperature of the shape (250–700 nm)	6489 K	of order 10^{4} K and above	too cool

Time-integrated spectrum (energy per nm, relative): 250 nm 0.55, 325 nm 0.81, 400 nm 0.95, 475 nm 1.00, 550 nm 0.99, 625 nm 0.95, 700 nm 0.90. An optically thin gas at the model's 15,000 K peak emits most in the blue-green. The processes left out (electron–neutral bremsstrahlung, recombination, heat loss, the lowered ionization potential) are where the standard account finds the ultraviolet. None of the shortfall points to a non-thermal source. The light is far below the other ledger terms and is not taken out of the dynamics.

Part 5: does the flash carry the knot's topology?

A knot of field energy carries net optical helicity; thermal light from random currents carries none. The bench measurement is the whole flash's degree of circular polarization, summed over every frequency and over 600 probes on a sphere (V > 0 is spin along the direction of travel):

```
P_c = ΣV/ΣI,  V = −2 Im(Ê_θ*·Ê_φ),  I = |Ê_θ|² + |Ê_φ|²
```

The sources run in em/fdtd.go, scaled to a central wavelength λ_0 = 450 nm at 8 cells: the collapsed core (R_{min} ≈ 0.79 μm) is 1.75 λ_0, and the probes sit at 6 λ_0.

Source	P_c	Field ledger residual	Energy absorbed
hopfion of knotted_light (exact field at the probes +0.9999)	+0.9996	1.6×10^{-13}	0.937
chiral dipole pair (p || m, quadrature)	+0.9797	5.7×10^{-15}	0.992
rotating electric dipole	+0.0007	4.3×10^{-15}	0.992
thermal core, seed 1	+0.0087	1.2×10^{-15}	0.999
thermal core, seed 2	−0.0015	2.9×10^{-15}	0.999

© 2026 fdamador@comcast.net

Source	P_c	Field ledger residual	Energy absorbed
thermal core, seed 3	−0.0268	3.5×10^{-14}	0.999
thermal core, seed 4	+0.0296	1.3×10^{-14}	0.999
thermal core, seed 5	+0.0069	1.3×10^{-14}	0.999
thermal core, seed 6	+0.0176	4.1×10^{-15}	0.999
thermal core, 6 seeds	**mean +0.0058, rms 0.0184**	**≤ 4×10^{-14}**	

- **The knot is one helicity.** Rañada's hopfion in the Bateman form satisfies i∂F/∂t = curl F, the condition for one helicity; the measurement sees +0.9996 and the exact field +0.9999.
- **An electric dipole alone cannot carry helicity**, even when circular on its axis: the two hemispheres cancel. Helicity needs electric and magnetic parts together, being the charge of electric–magnetic duality (Calkin, *Am. J. Phys.* 33, 958, 1965; Cameron, Barnett & Yao, *New J. Phys.* 14, 053050, 2012). Random hot-gas currents are electric currents.
- **The thermal core gives P_c consistent with zero**: rms 0.0184 against the independent-radiator estimate $1/\sqrt{(2N)} = 0.0408$. A real flash of 10^5–10^7 photons has a counting floor $1/\sqrt{N}$ of 3×10^{-3} to 3×10^{-4}.

The interactive report (viz/report.html) shows these results as browser-drawn charts and 3-D scenes: the bubble beside the hopfion over one cycle, the radius and ledger curves, the polarization globe and field slices. Those charts are generated in the browser from embedded data and have no image files, so this PDF gives their numbers in the tables above.

Validation

Test	Checks	Held to
TestRayleighPlessetLimit	lossless circuit = Rayleigh–Plesset	10^{-9}
TestLinearAdmittance	small-amplitude Y_1 = linear acoustic admittance	10^{-4}
TestCycleLedgerAndTwin	ledger; $\frac{1}{2}\mathrm{Re}(Y_1)P_a^2$ = cycle work × f; Keller–Miksis R_{max}, R_{min}, phase; flash plausible	10^{-8}; 10^{-6}; 1 %, 10 %, 0.5 %
TestChainWithBubble	bubble element takes $\frac{1}{2}\mathrm{Re}(Y_1)P_a^2$, mode heat unchanged, kronspice ledger	10^{-3}; 10^{-9}
TestReversal	lossless return, lossy failure	10^{-9}; > 10^{-2}
TestStokesSign, TestIonization	sign of V; Saha limits	exact
TestHelicity	hopfion = exact, chiral > 0.9, rotating and thermal ≈ 0, ledgers	0.01; 10^{-10}
em.TestFDTDSharedEdgeSources	two sources on one edge act as their sum, ledger closed (the solver fix this example needed)	10^{-10}

Limits of the model

- The circuit is first order in Ṙ/c; near the collapse neither it nor Keller–Miksis is exact (wall Mach 0.781 and 0.516).
- The light is the simplest standard estimate: uniform, optically thin core with Saha ionization and electron–ion free–free emission only. It is low by one to two orders in photons and too red.
- The polarization test uses scaled field-solver sources (λ_0 = 450 nm at 8 cells), 300 random current elements for the thermal core and six seeds.
- Part 2 is a harmonic balance at the fundamental: exact for power, not for the waveform.

Verdict and status

- **Where the analogy holds:** the bubble is a driven implosion and explosion with its inward supply booked, one cycle per acoustic period: 14.3 nJ in per cycle from the flask's standing wave and 13.3 nJ out as a sound pulse. This is the structure H30 found for the pulsing knot, with the flask's standing wave as the declared medium.
- **Where it fails:** the collapse is a nonlinear, irreversible flow of matter, stopped by gas pressure; its flow is spherically symmetric and carries no topology. The light is thermal in every test made: its width does not depend on colour, and its polarization is zero within the speckle of random currents, against +0.9996 for the knot.

© 2026 fdamador@comcast.net

- **Bench test:** a polarimeter on single flashes. A knotted field would give P_c of order 1; thermal light gives at most the counting floor. The two are separated by about three orders of magnitude.

H30 is recorded as *bounded* in the register; this example is its bench case, the sonoluminescing bubble as the same driven cycle with its inward supply booked.

Reproducing the results

```
go run ./examples/bubble_collapse            # about 1 min
go run ./examples/bubble_collapse -nofield   # parts 1–4 only
go test ./examples/bubble_collapse           # -short skips the field solver
go run ./examples/bubble_collapse -data report_data.json
go run ./examples/bubble_collapse -report examples/bubble_collapse/viz/report.html
python examples/bubble_collapse/viz/make_report.py examples/bubble_collapse/bubble_collapse_report.pdf
```

Sources: Brenner, Hilgenfeldt & Lohse, *Rev. Mod. Phys.* 74, 425 (2002); Rayleigh, *Phil. Mag.* 34, 94 (1917); Plesset, *J. Appl. Mech.* 16, 277 (1949); Keller & Miksis, *J. Acoust. Soc. Am.* 68, 628 (1980); Kinsler et al., *Fundamentals of Acoustics*, 4th ed., §7.2; Kron (1951), Epilogue eqs. 3–4; Hilgenfeldt, Grossmann & Lohse, *Nature* 398, 402 (1999); Rybicki & Lightman (1979); Gompf et al., *PRL* 79, 1405 (1997); IAPWS R1-76(2014), R12-08; Calkin (1965); Cameron, Barnett & Yao (2012); Rañada (1989); Kedia et al., *PRL* 111, 150404 (2013).

© 2026 fdamador@comcast.net